

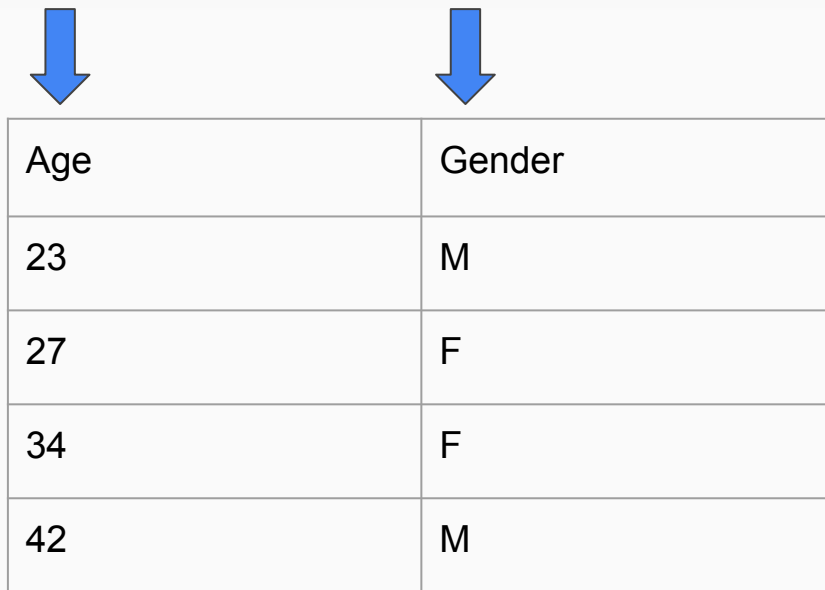
Parquet

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Parquet can improve performance.

- Column based storage
- Reduced Input/Output
- Compression
- Metadata

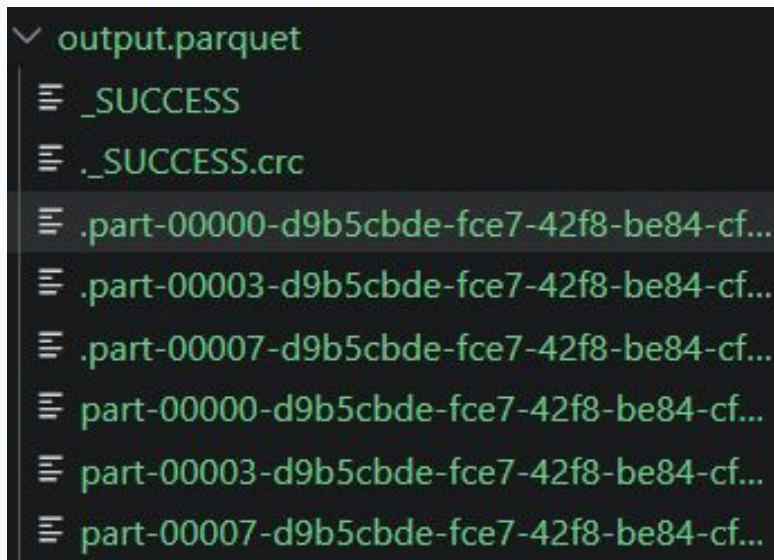


The diagram illustrates columnar storage. Two blue arrows point downwards to the headers of two columns in a table. The first column is labeled 'Age' and the second is labeled 'Gender'. Below the headers, the data is organized into rows: 23, 27, 34, 42 for Age and M, F, F, M for Gender.

Age	Gender
23	M
27	F
34	F
42	M

Parquet has several use cases.

- Data Lakes
- When queries only need a few columns
- If data is written as a batch
- Efficiency matters more than readability



```
output.parquet
├── _SUCCESS
├── _SUCCESS.crc
├── .part-00000-d9b5cbde-fce7-42f8-be84-cf...
├── .part-00003-d9b5cbde-fce7-42f8-be84-cf...
├── .part-00007-d9b5cbde-fce7-42f8-be84-cf...
├── part-00000-d9b5cbde-fce7-42f8-be84-cf...
├── part-00003-d9b5cbde-fce7-42f8-be84-cf...
├── part-00007-d9b5cbde-fce7-42f8-be84-cf...
```

Parquet has limitations.

- Not optimized for frequent updates
- Large datasets should be partitioned first
- Not optimized for many small files
- Complex schema management



Resources

- <https://www.datacamp.com/tutorial/what-is-a-parquet-file>
- <https://spark.apache.org/docs/latest/sql-data-sources-parquet.html>
- <https://www.sparkcodehub.com/pyspark/data-sources/write-parquet>
- <https://chatgpt.com/>